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Investigating Gender Bias in BERT

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Abstract
In this work, we analyze the gender bias induced by BERT in downstream tasks. We also propose solutions to reduce gender bias. Contextual language models (CLMs) have pushed the NLP benchmarks to a new height. It has become a new norm to utilize CLM-provided word embeddings in downstream tasks such as text classification. However, unless addressed, CLMs are prone to learn intrinsic gender bias in the dataset. As a result, predictions of downstream NLP models can vary noticeably by varying gender words, such as replacing "he" to "she", or even gender-neutral words. In this paper, we focus our analysis on a popular CLM, i.e., BERT. We analyze the gender bias it induces in five downstream tasks related to emotion and sentiment intensity prediction. For each task, we train a simple regressor utilizing BERT's word embeddings. We then evaluate the gender bias in regressors using an equity evaluation corpus. Ideally and from the specific design, the models should discard gender informative features from the input. However, the results show a significant dependence of the system's predictions on gender-particular words and phrases. We claim that such biases can be reduced by removing gender-specific features from word embedding. Hence, for each layer in BERT, we identify directions that primarily encode gender information. The space formed by such directions is referred to as the gender subspace in the semantic space of word embeddings. We propose an algorithm that finds fine-grained gender directions, i.e., one primary direction for each BERT layer. This obviates the need of realizing gender subspace in multiple dimensions and prevents other crucial information from being omitted. Experiments show that removing embedding components in gender directions achieves great success in reducing BERT-induced bias in the downstream tasks. The investigation reveals significant gender bias a contextualized language model (i.e., BERT) induces in downstream tasks. The proposed solution seems promising in reducing such biases.

Keywords Gender bias · BERT · Contextual language models · Natural language processing.

Introduction

Gender stereotypes can obstruct gender neutrality in many areas such as education, work, politics. Despite years of headway towards gender neutrality, the significant bias in social norms still exists. Automatic machine learning systems are likely to reproduce and reinforce existing gender stereotypes. Such issues have percolated down to even the language models that have recently set the state of the art in various natural language processing (NLP) tasks [1]. However, the blunt application of language models risks introducing gender bias in real-world systems.

It is becoming increasingly common to use an LM's contextualized word-vectors in downstream tasks such as text classification, question-answering, and conference resolution. In this work, we focus our analysis on one of the most famous language models: Bidirectional Encoder Representations for Transformers known as BERT [2]. BERT is a transformer-based architecture [3] that has inspired many recent advances in machine learning even beyond language-only systems [4]. BERT allows parallelized training and deals with long-range dependencies better than RNN-based models such as ELMo [5, 6].

Existing studies mostly focus on identifying gender bias in context-independent word representations such as GloVe [7]. Contrarily, BERT word to vector(s) mapping is highly context-dependent which makes it difficult

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to analyze biases intrinsic to BERT. We hypothesize that such biases will be reflected in downstream tasks exploiting BERT word embeddings. Hence, in this work, we investigate gender bias induced by BERT in 5 downstream tasks that collectively fall in a category of tasks–Affect in tweets. The category splits into two sub-categories 1) emotion intensity and 2) valence (sentiment) intensity regression.

To perform the above-mentioned tasks, we train simple MLP-regressors exploiting BERT embeddings. We probe gender bias in the trained models using an equity-evaluation corpus. The corpus consists of sentences especially designed to tease-out biases in NLP systems. Ideally, MLPs should not base their predictions on gender-specific words or phrases in the input. However, we observe the MLPs to consistently assign higher (or lower) scores to the sentences with words or phrases indicating a particular gender. For instance, one of the MLP regressors predicts high emotion intensity scores to sentences with female words than male words under the same context. We call such systems as **gender-biased**. It is worth noting that the gender inclination is found to be specific to the task and word embedding used, hence ungeneralizable.

Due to the simplicity of MLP's, we hold BERT accountable for the observed gender bias. Subsequently, we show the existence of layer-specific orthogonal directions where BERT encodes crucial gender information. We call such direction as **gender directions** and the space spanned by them as **gender subspace**. The directions (thus subspace) is unique to a BERT layer. The quality of extracted gender directions is identified by defining a new metric **gender separability**. To reduce the number of dimensions of gender subspace, we propose a novel algorithm that identifies fine-grained gender directions, i.e., one for every BERT layer. Thus, the obtained gender subspaces are 1-dimensional. The layer-wise elimination of vector components in gender directions helps reduce gender bias in the downstream regression models.

To establish the importance of extracted gender directions, we design another downstream task, i.e., gender classification. The task specifically needs gender encoded features from the input word's vector representation. We find the BERT-based MLP to outperform a baseline gender classifier, proving the existence of gender-rich features in BERT embeddings. Additionally, removing gender-specific directional components from BERT embeddings drops the classification performance significantly. This concludes that the identified directions are close to the directions in BERT embeddings that encode the notion of gender.

Ethical concerns—In this work, the protected attributes being discriminated against are binary. The work is generalizable for genders being a non-binary attribute.

Related Work

While a lot has been studied, identified, and mitigated when it comes to gender bias in static word embeddings [7–10], very few recent works studied gender bias in contextualized settings. We adapt the intuition of possible gender subspace in BERT from [7], which studied the existence of gender directions in static word embeddings. [11–13] focused their study on ELMo. [14] provided a template-based approach to quantify bias in BERT. [15] studied bias in both contextualized and non-contextualized Swedish embeddings. Concurrent to our study, [16] mitigates the bias by nulling out the projection on hyperplane learned by a linear classifier using word embedding.

In the literature of attribute probing, many recent advances happened. An information-theoretic probing was introduced in [17], which measures minimum description length (MDL) of labels given representations. [18] put probe accuracy in context with its ability to remember from word types. Amnesic probing [19] observes the change in the prediction of the classifier caused by the removal of property Z from the representation of the input. It does so by learning a classifier to predict the property. In this work, we adapt probing strategy from [20, 21] for the regression tasks as our approach does not rely on learning a gender classifier, however, we use it to evaluate gender debiasing.

To the best of our knowledge, we are the first to identify gender bias in BERT by analyzing its impact on downstream tasks. We propose a novel algorithm to identify fine-grained gender directions to minimize the exclusion of important semantic information. Empirically, the elimination of embedding components in gender directions proves to be significantly reducing gender bias in the tasks under study.

Background

BERT In our study, we analyze BERT base–12 layers (transformer blocks), 12 attention heads, and 110 million parameters. The model is pre-trained on masked-language model and next sentence prediction tasks [2] on lower-cased English text. Out-of-vocabulary (OOV) words are WordPiece tokenized that breaks a word into subwords from the predefined vocabulary. An input sequence of words W is prepended with `[CLS]`, appended with `[SEP]`, and tokenized to generate $W_t = \{w_{cls}, w_1, ..., w_n, w_{sep}\}$. First, tokens are mapped to context-independent vectors $W_0 = \{t_0^{cls}, t_0^0, \ldots, t_0^n, t_0^{sep}\}$, we denote it as $layer_0$. $\{layer_i\}_{i=1}^{12}$ are transformer layers that map vectors in W_0 to contextualized vectors $\{W_i\}_{i=1}^{12}$. We denote $t_j^i \in \mathbb{R}^{d_b}$ as vector representation of w_j at the output of $layer_i$. We utilize vocabulary and BERT pre-trained model from [22].

Equity Evaluation

Equity Evaluation Corpus (EEC) [23] The dataset contains template-based sentences such as "<Name> feels angry". <Name> can be a female name such as "Jasmine", or a male name such as "Alan". An NLP-model is then asked to predict the intensity of emotion - angry. A system is called gender-biased when it consistently predicts higher/lower scores for sentences carrying female-names than male-names, or vice versa. The EEC contains 7 templates of type: <person> and <emotion>. The place of variable <person> can be filled by any of 60 gender-specific names or phrases. Out of 60, 40 are gender-specific names (20-female, 20-male). Rest 20 are noun phrases, particularly, 10 female-male pairs such as "my mother" and "my father". Variable <emotion> can replace four emotions – Anger, Fear, Sadness, and Joy, each having 5 representative words [1]. Thus, there are 1200 ($60 \times 5 \times 4$) samples for each template. Therefore, there are 8400 (7×1200) samples equally divided in female and male-specific sentences ($60 \times (5 \times 4) \times 7 = 4200$ each) and 4-emotion categories ($5 \times 7 \times 60 = 2100$ each). We refer readers to [23] for an elaborate explanation.

Evaluation Methodology To evaluate an NLP system for its intrinsic gender bias, we follow the same evaluation scheme as in [23]. For a given template T and emotion word E, i.e., T-E format in EEC, we obtain 11 pairs of female-male intensity scores. One of the pairs is obtained by averaging the system's intensity predictions of input sentences with gender-specific names. The score pair consists of an average female score as its first element and a male score as its second. The other 10 scores are calculated from 10 noun phrase pairs used in the same T-E format. Thus for 7 templates and 20 emotion words, we have $7{\times}20{\times}11 = 1540$ pairs of scores. We define $\Delta_{F\uparrow-M\downarrow}$ as the average difference in female to male scores for those pairs with higher predicted intensity for females; vice versa to this defines $\Delta_{M\uparrow-F\downarrow}$. The number of occurrences where female scores are higher ($\#_{F\uparrow-M\downarrow}$), male scores are higher ($\#_{M\uparrow-F\downarrow}$), and both female-male scores are equal ($\#_{F=M}$) are also kept for a fine-grained evaluation.

BERT Induced Bias

As mentioned earlier, we hypothesize that downstream tasks are prone to acquire gender bias from BERT word embeddings. However, it is possible that a task-specific model enhances or diminishes the BERT induced bias, or learns its own bias. To prevent such scenarios, for all the tasks, we use shallow MLP regressors without fine-tuning BERT parameters. The simplicity of regressors will expose inherent gender bias in BERT. Next, we elaborate on the downstream tasks, the MLP architecture, and the training procedure.

Downstream Tasks

Our bias evaluations are based on regression subtasks of SemEval-2018 Task 1–Affect in Tweets [24]. The sub-tasks are majorly classified in:

1. Emotion intensity regression (E_R-tasks): Given a tweet and an affective dimension $\in$ {joy, fear, sadness, anger}, determine emotion intensity I–a real-valued score between 0 (low mental state) and 1 (high mental state).
2. Sentiment intensity regression (S_R-task): Similar to E_R tasks, given a tweet, determine intensity I of the sentiment.

For all the five tasks, i.e., 4-E_R tasks and an S_R task, train and test sets are provided with gold intensity scores.

BERT-based MLP Regressor (BERT-MLP)

Since the `[CLS]` token was specifically introduced as a representative of the input sequence, it is reasonable to use its vector representations in downstream tasks (shown in Fig. 1). For the sequence W, we obtain a deep contextualized vector representation of `[CLS]`, i.e., W_k^{CLS} from $layer_k$[2].

For each task, we train a 2-layer MLP regressor to predict intensity $I_k \in [0, 1]$ expressed by the sequence W, i.e., a tweet. Hence we train-test 5 different regressors. Input to the MLP is $W_k^{CLS} \in \mathbb{R}^{768}$, hidden layer carries 200 neurons fully connected to the input vector, followed by Relu activation. The output is just an affine combination of the values obtained after activation. The Adam optimizer minimizes squared-loss between the MLP outputs and ground-truth intensity values. We divide the dataset into batches of 200 samples. In each iteration, parameters are updated to reduce loss accumulated in one batch. We score the models by calculating Pearson's correlations between predicted and expected intensity values[3]. The MLP architecture is kept simple to decipher features encoded in BERT word representations. Thus, the task performance of an MLP will rely heavily on the features provided by BERT. BERT columns in Tables 2 and 4 show Pearson scores. We restrict our analysis to embeddings from deep layers in BERT i.e., $layer_{11}$ and $layer_{12}$.

[1] e.g.,:- {angry, enraged} represents a common emotion, i.e., anger

[2] W_k^{CLS} and t_k^{cls} represent the same vector.

[3] Following https://competitions.codalab.org/competitions/17751#results

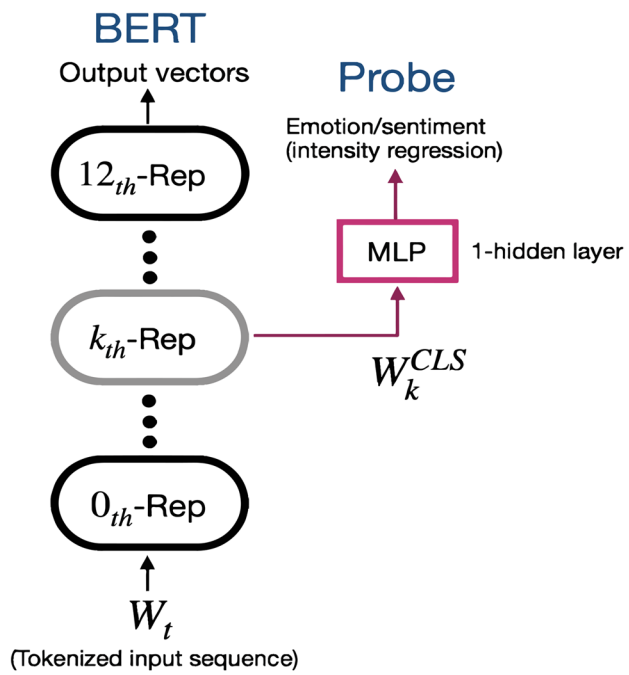

Fig. 1 MLP regressor trained on five individual regression tasks. Input to the MLP is 768-dimensional vector mapping of [CLS] token at BERT $layer_k$

Equity Evaluation of BERT-MLP

We evaluate $layer_{11}$ and $layer_{12}$ embedding of BERT separately. As shown in the Tables 2 and 4, in columns correspond to BERT, all five regressors show significant Δ values[4]. For each task, we observe that the regressors consistently assign high values to either of the genders. Moreover, not many cases are seen where MLP's assign equal scores to both the genders, i.e., $\#_{F=M}$. We discuss the results in later sections.

Gender Debiasing

In this section, we aim to uncover principal directions where BERT layers encode gender information. We hypothesize that removing the word vector components from gender directions will lead to reduced gender bias in downstream tasks utilizing BERT embeddings.

1. Independently for each BERT layer, we find a gender direction that encodes gender information.
2. We evaluate the quality of obtained directions by defining a new metric gender separability.
3. Subsequently, we propose Algorithm-1 to obtain fine-grained gender directions and to introduce a new setting–BERT^{De}–which lacks in gender-rich features.

Following [7] work on identifying gender axis (direction) in context-independent word embedding, we extend it to extract geometric directions from contextualized word embeddings of BERT. We hypothesize–*For every layer in BERT, there exists a low-dimensional context-independent subspace that encodes gender information.*

Thus, for each BERT layer $layer_k$ $(0 \leq k \leq 12)$, we aim to capture a d-dimensional gender subspace B_k spanned by the basis vectors $\{b_k^1, \ldots, b_k^d\} \in \mathbb{R}^{d_b}$. We define the basis-vectors as *gender directions*. Ideally, the difference in vector representations of He and She should show a major component in gender directions. However, even in simple static embedding like Glove, such vectors may not behave as expected [7]. The task of revealing gender subspace becomes even more difficult in case of contextualized embeddings, i.e., a word can map to more than one vector representations depending on the context. Such embeddings may lead to inconsistency in extracted directions, and thus, subspace. We propose a way to identify a static gender subspace B_k by enforcing context to have many gender-specific words, all of which represent the same gender. We further elaborate on the method below.

Definition pair Let $O_g := \{(f_i, m_i)\}_{i=1}^{g}$ be the ordered pair of words. The f_i represents a noun that is commonly used for a female. Similarly, m_i carries a male notion [5]. Using O_g, we form a definition pair of sentences:

$$S_f = w_1 \ldots f_1 \ldots f_g \ldots w_n$$
$$S_m = w_1 \ldots m_1 \ldots m_g \ldots w_n$$

The definition pair makes use of the O_g in a close context. We denote word at position i in sequence S_f and S_m as S_f^i and S_m^i. The definition set (S_f^i, S_m^i) satisfies either of the two conditions:

1. (S_f^i, S_m^i) is gender pair in O_g; or
2. $S_f^i = S_m^i = w_i$, where w_i is a gender-neutral word.

From 10 gender pairs introduced in [7], we chose 9 and added {Queen,King} and {Aunt,Uncle}. Thus, O_g contains 11 gender pairs. Our experimental findings suggest that more number of gender pairs make it difficult to find principal directions. As mentioned before, we use O_g along with gender-neutral words to generate S_f and S_m.

Let u_k^i and v_k^i denote vector mapping of words S_f^i and S_m^i at BERT $layer_k$, respectively. Since S_f and S_m are the same

[4] Delta values can be compared to models studied in [23]

[5] We focus on those words having low word-sense ambiguity

except for gender-specific words, we expect their word vectors to have a close contextual relationship. Thus, we conjecture that the difference vector $D_k^i = \{v_k^i - u_k^i\}$ shows a noticeable shift in gender directions by canceling out other encoded information such as context and word position. Later, we empirically show the importance of gender directions extracted using difference vectors.

Gender Subspace

Independently for each $layer_k$, principal component analysis (PCA) over difference vectors $D_k := \{D_k^i\}_{i=1}^n$ helps uncover gender subspace B_k. $\texttt{PCA}(D_k)$ returns n-orthogonal directions $p_k^1, \ldots, p_k^n$ in decreasing order of the explained variance (EV). Each direction encodes a different notion that supplements the abstract concept of gender. Higher EV signifies more crucial directions that define gender subspace. However, such directions may not behave as expected and encode other information unrelated to gender. Initially, we base our analysis on two principal directions, i.e., p_k^1, p_k^2 forming the 2-dimensional gender subspace. Subsequently, we show both the gender directions have a significant overlap in the encoded information. This gives an intuition of primarily using the first principal component that leads to a 1-dimensional subspace for gender Fig. 2.

Gender Separability Consider a direction l and vector representation of words g_1, g_2 as x and y, where g_1 is a masculine word and g_2 is feminine, or vice versa. We can find a real value c such that:

$$\langle x, l \rangle \geq c \text{ and } \langle y, l \rangle \leq c \tag{1}$$

c divides l in two rays (half spaces) each of which represents a unique gender. An ideal gender direction should project a word-vector in gender-specific ray. Thus, we define gender-separability as the accuracy of projection-based classification on the corresponding ray.

Gender Classification Dataset (Gen-data) We compiled train-set from [25] [6]. The dataset consists of 222 gender-word pairs, i.e., for each feminine word, there is a masculine counterpart. To form test-set, we collect gender-specific words from another source containing 595 male-specific and 404 female-specific words [7]. From this source, we collected 595 neutral words and randomly split them to assign 222 samples for training and rest for testing. Table 1 shows Gen-data statistics and samples.

We evaluate layer-specific gender subspaces for their gender separability on Gen-Data, we evaluate the first two principal directions p_k^1 and p_k^2 of a given $layer_k$. To find

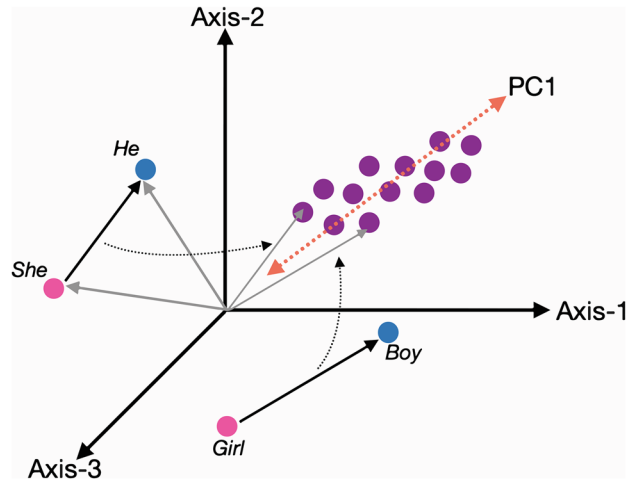

Fig. 2 Illustration of capturing the first principal component using gender pairs in a three-dimensional word embedding space

layer-specific c, i.e., c_k^i for i_{th} principal component (PC), we perform a grid search to maximize separability on the train-set of Gen-data excluding gender-neutral words. We then test the quality of separation on its test-set as shown in the Fig. 3. We observe the first PC of all layers have high separability score. Moreover, second PCs of middle layers performs as good as respective first PCs. This observation raises a question:- is second gender direction, i.e., p_k^2 crucial to define the gender subspace? We answer it by the following analysis:

Since all the contextual embeddings are transformations of embedding at $layer_0$, in Fig. 4, we plot the cosine similarity between first principal gender direction of $layer_0$ (p_0^1) and $layer_j$ (p_j^1), where $1 \leq j \leq 12$. The high cosine similarity [8] depicts passing of encoded gender information from $layer_0$ and lesser new gender-specific features learned by following layers. We also analyze cosine similarity between p_0^1 and p_j^2 which is the second principal gender direction at $layer_j$. The similarity score increases in the middle layers which supports our observation in Fig. 3 with high separability values. Moreover, it also indicates that p_j^2's hardly encode any extra gender-specific information keeping aside what is acquired from p_0^1.

Eliminating vector components in gender directions is expected to reduce gender bias in downstream tasks. However, due to non-ideal behavior, omitting a large number of directional components may cause representation noise and hinder the quality of BERT embeddings. Moving forward, we propose Algorithm-1 that aims to tackle this issue.

[6] https://github.com/uclanlp/gn_glove/tree/master/wordlist

[7] https://github.com/ecmonsen/gendered_words

[8] cosine of the angle between two random vectors in high dimensions is zero with high probability.

Table 1 Distribution of Gen-data

Data	Gender	#Samples	Examples
Train	Female	222	actress,mama,madam, princess,sororal
	Male	222	actor,papa,sir, prince,fraternal
	Neutral	222	guest, beast, friend mentor, outlier
Test	Female	404	chatelaine, ballerina, baroness, barmaid, brunette
	Male	595	adonis, barman,baron, brunet, charon
	Neutral	5701	abator, owner, bidder, genius, whistler

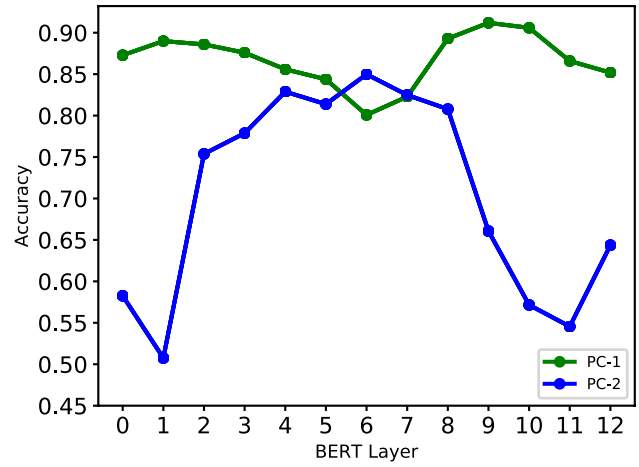

Fig. 3 Layer-wise gender separability on Gen-data when word vectors are projected on first principal component (PC-1) and second principal component (PC-2)

Reducing Gender Bias

For each layer in BERT, Algorithm-1 extracts first principal gender directions p_j^1 (referred as P_j) in a systematic way. The $layer_0$ maps WordPiece tokens of S_f and S_m to a set of vectors u_i and v_i, respectively. PCA over difference vector $D_0 = (v_0 - u_0)$ gives P_0 i.e. gender direction with maximum explained variance. We remove components of v_0 and u_0 on P_0 by taking perpendicular projections. For a vector a, the projection perpendicular to a unit vector b is defined as:

$$Proj_{\perp b}(a) := a - \langle a, b \rangle b \tag{2}$$

(Where $\langle a, b \rangle$ is inner product of vectors a and b.)

$P := P_1, \ldots, P_{12}$ are stored. It is worthwhile pointing that the algorithm is different from independent layer-wise analysis as each layer has missing gender information from layers preceding it. The new cosine similarity scores show a significant drop in Fig. 4 - dotted.

Removing Gender Component After obtaining layer-wise gender directions P, we introduce a new BERT setting–BERT^{De}. As an enhancement of BERT, BERT^{De} removes gender components from a token's vector representations. Given an input sequence of tokens W_t to the $layer_0$,

ALGORITHM 1: Extracting layer-wise principal component in Gender subspace.

Input :- Strings pair (S_m, S_f), which differ only in gender-specific words.
Output :- P=Layer-wise principal component set $\{P_0, \ldots, P_{12}\}$.

$W_{tf} \leftarrow \texttt{Tokenize}(S_f)$ /* WP Tokenization */
$W_{tm} \leftarrow \texttt{Tokenize}(S_m)$ /* WP Tokenization */
$u_0 \leftarrow \texttt{Layer}_0(W_{tf})$ /* Context-independent input vectors for S_f */
$v_0 \leftarrow \texttt{Layer}_0(W_{tm})$ /* Context-independent input vectors for S_m */
$D_0 \leftarrow (v_0 - u_0)$ /* Difference vector */
$P_0 \leftarrow \texttt{PCA}(D_0)$ /* PC with maximum EV */
for $j \leftarrow [1, 2, \ldots, 12]$ **do**
  $u^*_{j-1} \leftarrow \texttt{Proj}_{\perp P_{j-1}}(u_{j-1})$ /* Perpendicular projection */
  $v^*_{j-1} \leftarrow \texttt{Proj}_{\perp P_{j-1}}(v_{j-1})$
  $u_j \leftarrow \texttt{Layer}_j(u^*_{j-1})$
  $v_j \leftarrow \texttt{Layer}_j(v^*_{j-1})$
  $D_j \leftarrow (v_j - u_j)$ /* Difference vector */
  $P_j \leftarrow \texttt{PCA}(D_j)$
end

We feed the projected vectors v_0^* and u_0^* to the next layer, i.e., $layer_1$. The same procedure is repeated until final layer $layer_{12}$ and all the extracted principal components we obtain token-vectors W_0 at its output. For each vector t_0^i in W_0, we remove its component in direction P_0, i.e., $t_0^{i*} = Proj_{\perp P_0}(t_0^i)$. We denote the set of t_0^{i*} vectors as W_0^*.

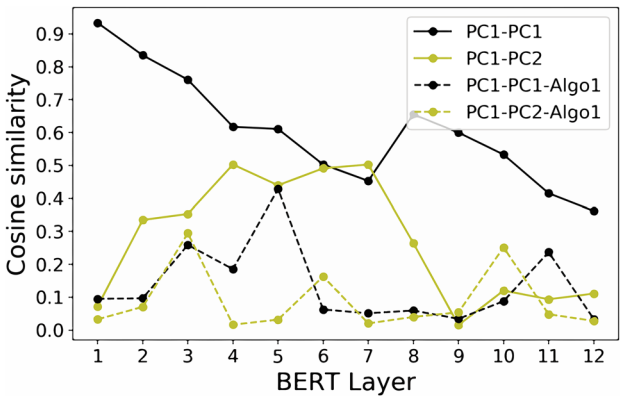

Fig. 4 Cosine similarity between two vectors $Cos(u, v) = \frac{u \cdot v}{||u||||v||}$. PC1-PC1, PC1-PC-2 denotes $Cos(p_0^1, p_j^1)$ and $Cos(p_0^1, p_j^2)$, respectively. Algo1 denotes PCs obtained using Algorithm 1

Unlike normal BERT settings which feed W_0 as input to $Layer_1$, we feed W_0^*. We iterate this process for every layer $layer_j$ ($0 \leq j \leq 12$) which receives W_{j-1}^* ($j > 0$) at input and gives W_j^* at output by removing its vector components in direction P_j. In the next section, we evaluate BERT^{De} on EEC and compare its performance with BERT.

Equity Evaluation of BERT^{De} - MLP

We follow the same methodology as in Section 5.2, however, by substituting BERT with BERT^{De}. For each task-specific MLP trained on BERT^{De} $layer_{11}$ and $layer_{12}$, we perform paired two-sample t-tests to determine whether the mean difference between male and female scores is significant. Low p-values indicate a significant difference in model predictions based on gender.

Results and Discussion

As shown in the Tables 2 and 4, most of the BERT^{De}-MLP regression models show an overall % decrease in Δ values in both $F>M$ ($\Delta_{F\uparrow-M\downarrow}$) and $M>F$ ($\Delta_{M\uparrow-F\downarrow}$) cases. Final-layer BERT-MLP models for joy, fear, and anger have higher average intensity scores for male phrases than female, while the opposite trend is seen in models for valence and sadness. It is also noteworthy that the pre-final layer ($layer_{11}$) shows a somewhat opposite trend. Hence, we suspect the BERT -induced bias depends on the which layer embedding is used. Moreover, from p-values of BERT-based regressors, we see much higher significant Δ values as compared to regressors using BERT^{De}. From Tables 3 and 5, for all five regressors, we observe a significant reduction in difference between number occurrences where $F>M$ and $M>F$, i.e., δ. We also

Table 2 Final-layer ($layer_{12}$) of BERT and BERT^{De} equity evaluation of the five-intensity regression models. %d refers to the percentage change in Δ values. The p-values for 1) Emotion intensity models: {anger}≤ 0.05(≥0.70*); {joy, fear, sad} ≤ 0.20(≥0.70*). 2) The valence intensity model (emotion-wise p-values): {anger}≤ 0.05(≥0.75*); {joy, fear, sad}≤ 0.20(≥0.70*), where values with * denotes BERT^{De}-based MLP regressor

Emotion	Emotion Intensity						Valence Intensity					
	BERT			BERT^{De}			BERT			BERT^{De}		
	Pearson	$\Delta_{F\uparrow-M\downarrow}$	$\Delta_{M\uparrow-F\downarrow}$	Pearson	$\Delta_{F\uparrow-M\downarrow}$(%d)	$\Delta_{M\uparrow-F\downarrow}$(%d)	Pearson	$\Delta_{F\uparrow-M\downarrow}$	$\Delta_{M\uparrow-F\downarrow}$	Pearson	$\Delta_{F\uparrow-M\downarrow}$(%d)	$\Delta_{M\uparrow-F\downarrow}$(%d)
Joy	0.666	0.0396	0.0402	0.660	0.0143(↓63.9)	0.0143(↓64.4)	0.659	0.0346	0.0376	0.670	0.0209(↓39.5)	0.0138(↓63.3)
Fear	0.581	0.0202	0.0244	0.593	0.0152(↓24.7)	0.0158(↓35.2)		0.0263	0.0244		0.0156(↓40.6)	0.0123(↓49.5)
Sadness	0.615	0.0380	0.0138	0.604	0.0178(↓58.9)	0.0097(↓29.7)		0.0272	0.0205		0.0153(↓43.7)	0.0118(↓42.4)
Anger	0.627	0.0074	0.0316	0.626	0.0121(↑63.5)	0.0149(↓52.8)		0.0219	0.0198		0.0130(↓40.6)	0.0119(↓39.8)

Table 3 Final layer of BERT and BERT^{De} equity evaluation: $F\uparrow - M\downarrow$, $M\uparrow - F\downarrow$, and $M{=}F$. $\delta = |\#_{F\uparrow-M\downarrow} - \#_{M\uparrow-F\downarrow}|$

Emotion	Emotion Intensity								Valence Intensity							
	BERT				BERT^{De}				BERT				BERT^{De}			
	$\#_{F\uparrow-M\downarrow}$	$\#_{M\uparrow-F\downarrow}$	δ	$(\#_{F=M})$	$\#_{F\uparrow-M\downarrow}$	$\#_{M\uparrow-F\downarrow}$	δ	$(\#_{F=M})$	$\#_{F\uparrow-M\downarrow}$	$\#_{M\uparrow-F\downarrow}$	δ	$(\#_{F=M})$	$\#_{F\uparrow-M\downarrow}$	$\#_{M\uparrow-F\downarrow}$	δ	$(\#_{F=M})$
Joy	92	291	198	2	197	177	20(↓178)	11(↑9)	62	322	260	1	227	147	80(↓180)	11(↑11)
Fear	177	204	27	4	175	199	24(↓3)	10(↑6)	105	276	171	4	207	165	42(↓129)	13(↑9)
Sadness	339	44	294	2	243	128	114(↓180)	14(↑12)	106	274	168	5	209	162	47(↓121)	14(↑9)
Anger	18	366	347	1	161	212	52(↓295)	12(↑11)	126	258	132	1	229	148	81(↓51)	8(↑7)

Table 4 Pre-final layer ($layer_{11}$) of BERT and BERT^{De} equity evaluation of the five-intensity regression models. %d refers to the percentage change in Δ values. The p-values for 1) Emotion intensity models: {fear}≤ 0.01(≥0.95*); {anger, joy, sad} ≤ 0.20(≥0.85*). 2) The valence intensity model (emotion-wise p-values): {anger, fear}≤ 0.05(≥0.95*); {joy, sad}≤ 0.20(≥0.85*)

Emotion	Emotion Intensity						Valence Intensity					
	BERT			BERT^{De}			BERT			BERT^{De}		
	Pearson	$\Delta_{F\uparrow-M\downarrow}$	$\Delta_{M\uparrow-F\downarrow}$	Pearson	$\Delta_{F\uparrow-M\downarrow}$(%d)	$\Delta_{M\uparrow-F\downarrow}$(%d)	Pearson	$\Delta_{F\uparrow-M\downarrow}$	$\Delta_{M\uparrow-F\downarrow}$	Pearson	$\Delta_{F\uparrow-M\downarrow}$(%d)	$\Delta_{M\uparrow-F\downarrow}$(%d)
Joy	0.580	0.0436	0.0152	0.557	0.0195 (↓55.2)	0.0165(↑8.75)	0.658	0.0356	0.0118	0.653	0.0118(↓66.6)	0.0086(↓26.65)
Fear	0.475	0.0256	0.0241	0.497	0.0139(↓45.4)	0.0130(↓ 45.6)		0.0348	0.0113		0.0117(↓66.2)	0.0099(↓11.8)
Sadness	0.532	0.0282	0.0129	0.535	0.0156(↓44.4)	0.0133(↑2.8)		0.0192	0.0089		0.0185(↓3.38)	0.0113(↑26.6)
Anger	0.571	0.0123	0.0408	0.577	0.0133(↑8.6)	0.0124(↓ 69.4)		0.0185	0.0109		0.0177(↓4.24)	0.0115(↑5.57)

Table 5 Pre-final layer BERT and BERT^{De} equity evaluation: Number of occurrences of $F\uparrow - M\downarrow$, $M\uparrow - F\downarrow$, and $M{=}F$. $\delta = |\#_{F\uparrow-M\downarrow} - \#_{M\uparrow-F\downarrow}|$

Emotion	Emotion Intensity								Valence Intensity							
	BERT				BERT^{De}				BERT				BERT^{De}			
	$\#_{F\uparrow-M\downarrow}$	$\#_{M\uparrow-F\downarrow}$	δ	$(\#_{F=M})$	$\#_{F\uparrow-M\downarrow}$	$\#_{M\uparrow-F\downarrow}$	δ	$(\#_{F=M})$	$\#_{F\uparrow-M\downarrow}$	$\#_{\uparrow-F\downarrow}$	δ	$(\#_{F=M})$	$\#_{F\uparrow-M\downarrow}$	$\#_{M\uparrow-F\downarrow}$	δ	$(\#_{F=M})$
Joy	310	74	236	1	199	177	22(↓214)	9(↑8)	328	55	273	2	188	186	2(↓271)	11(↑9)
Fear	223	160	63	2	159	212	53(↓10)	14(↑12)	335	45	290	5	194	172	22(↓268)	19(↑14)
Sadness	281	99	182	5	210	169	41(↓141)	6(↑1)	283	85	198	17	202	173	29(↓169)	10(↑7)
Anger	31	352	321	2	145	225	80(↓241)	15(↑13)	269	109	160	7	217	155	62(↓98)	13(↑6)

observe an increase in cases when regressors assign equal scores to both genders, i.e., $\#F{=}M$. Unlike BERT, models based on BERT^{De} show no consistency in assigning higher intensity scores to either male or female. Hence, simple MLP regressors based on BERT^{De} vectors show an apparent gender unbiased nature on EEC.

Semantic Consistency

Gender debiasing of a model is desirable, however, it may come at the cost of reduced model performance on the task. In the case of BERT^{De}, removal of component in identified directions can lead to a loss of other semantic information. Thus, to check the semantic consistency of BERT^{De}, we compared Pearson's correlation score of BERT-MLP and BERT^{De}-MLP regressors on the task-specific test-sets. As depicted in the Tables 2 and 4, there is no drastic reduction in Pearson's scores, confirming semantic is preserved. Thus, removing the directional components reduces the gender bias induced by BERT, while maintaining the regressors performance on the downstream tasks. Next, we define a gender classification task to investigate the relevance of extracted directions, i.e., how informative they are about the gender of a given word.

The intuition behind Algorithm-1 is removing the layer-wise gender component from all the tokens. Obtaining gender directions does not rely on specific architecture or training setup. Thus, the algorithm is generalizable across contextual models such as RoBERTa [26], XLNet[27], and ELMo[5], irrespective of the adopted pre-training methodology. Any stereotypical gender association is expected to have a predominant component in one of the genders. Our method tries to eliminate the gender component from all the tokens. Hence, by default, all the gender associations will be neutralized.

Evaluation on Gender Classification

It is evident from the above analysis that BERT^{De} is effective in reducing gender-biased predictions of MLPs. Moreover, the semantic consistency proves BERT^{De} to be as effective as BERT on all the five tasks. To substantiate that Algorithm-1 makes BERT^{De} word embedding deficient in gender-specific features, we design another downstream task - gender classification of a word. A naive solution to reduce gender bias is to remove all gender-specific words. We analyze the suitability of this solution in the end.

Baseline Gender Classifier (GC)

First, we establish a gender classification baseline to compare the performances of BERT and BERT^{De}. Input to the baseline is WordPiece tokenized $w \rightarrow \{w_1, \ldots, w_N\}$. We randomly initialize the WordPiece embeddings - $E: w_i \mapsto t_i \in \mathbb{R}^{100} \sim \mathcal{U}(-1, 1)$. The possibility of multiple subwords makes it intuitive to perform convolution over t_i's [28]. The 1-dimensional convolution layer *Conv-1D* consists of 32 filters each of size 1. Thus the input tensor of shape $N \times 100$ after convolution at stride 1 transforms to $N \times 32$; this followed by global max-pooling gives 1×32 feature vector. The vector is passed through a fully connected layer consists of 128 neurons, and an output layer with *sigmoid* activation. We minimize the categorical cross-entropy of the output against the target gender set $\mathcal{L} \in \{0, 1\}^{|N|}$, where N: number of data samples, 1 and 0 are input labels for female and male, respectively (2-way classification). We use Adam optimizer with learning rate 0.001. We randomly drop-out 20% of *FC* layer activations to prevent parameter overfitting. Hyperparameters are tuned to maximize average 10-fold cross-validation accuracy.

MLP Gender Classifier

Similar to [29], we create a 2-layer MLP classifier. The architecture is very similar to Fig. 1 except for MLP is used for classification. The input to BERT is a word w. Given the output of BERT $layer_k$, i.e., $(t_k^{cls}, t_k^1, \ldots, t_k^n, t_k^{sep})$, we analyze two different input settings to MLP:

- I_1 : Vector representation of `[CLS]`, i.e., t_k^{cls}.
- I_2 : Average of all token vectors $t_k^{Avg} := \frac{t_k^{cls} + \sum_{i=1}^{n} t_k^i + t_k^{sep}}{(n+2)}$.

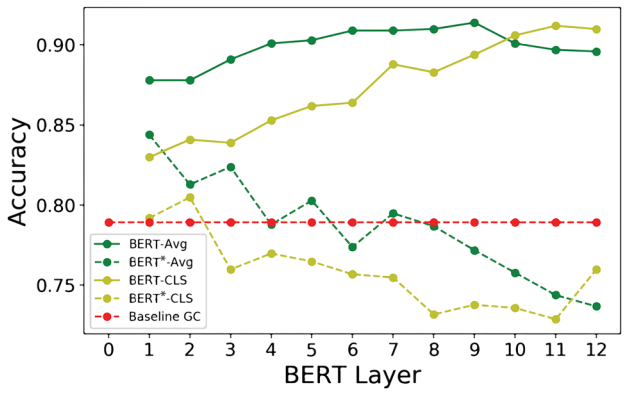

Fig. 5 BERT-CLS and BERT^{De}-CLS denote MLP accuracy using $layer_k$ (x-axis) vectors in I_1 setting. Similarly, BERT-Avg and BERT^{De}-Avg refer to the I_2 setting. Switching from BERT to BERT^{De}, we see a significant drop in MLP gender-classification performance in both I_1 and I_2 inputs cases

Table 6 Percentage of misclassified neutral words predicted as Male

	2-way	3-way	2-way	3-way
0	81.4	83.7	80.4(↓1.0)	81.4(↓2.3)
1	83.3	85.9	81.4(↓1.9)	81.6(↓4.3)
2	80.4	85.1	75.9(↓4.5)	79.0(↓6.1)
3	79.8	84.3	74.9(↓4.9)	83.1(↓1.2)
4	82.4	85.4	75.8(↓7.1)	82.3(↓3.2)
5	82.9	85.5	75.8(↓7.1)	81.2(↓4.3)
6	86.8	86.8	74.8(↓12.0)	81.2(↓5.6)
7	86.4	86.0	81.2(↓5.2)	79.1(↓6.9)
8	86.5	89.1	82.8(↓3.7)	70.9(↓18.2)
9	84.3	87.3	79.5(↓4.8)	71.9(↓15.4)
10	81.6	85.6	72.8(↓8.8)	74.7(↓10.9)
11	84.5	87.1	71.9(↓12.6)	74.8(↓12.3)
12	85.4	86.8	71.9(↓13.5)	67.9(↓18.9)
GC	Male: 51.1		Male: 50.3	
(Random)	Female: 48.9		Female: 49.7	

For a $layer_k$ and input setting, we train a separate MLP on the train-set of Gen-data and evaluate on its test-set. Each MLP takes I_1 or I_2 at input and predicts gender. Thus, given BERT model, we train-test 24 MLPs (12×2). The MLPs have 100 hidden layer neurons. We determine hyperparameters using a validation set comprised of the 10% samples from the training set. Rest settings are similar to the baseline.

BERT-MLP vs BERT^{De} - MLP

Following the above-mentioned method, we evaluate MLP classifiers based on BERT and BERT^{De} embeddings. As shown in the Fig. 5, we find BERT-based MLPs outperforming the gender classification baseline in both the setting I_1 and I_2. This depicts the existence of gender-rich features in BERT provided embeddings. BERT^{De}-MLP shows much poorer performance as compared to BERT-MLP. This observation makes it clear that removed directional components from the embeddings omit gender-rich features, hence, the obtained directions have a high magnitude of cosine similarity with actual gender directions. Moreover, BERT^{De}-MLP accuracy drops even below baseline at deeper layers, suggesting the similarity magnitude increases as the embeddings become deeply contextualized.

Additionally, we train MLPs on a 3-way classification task that includes gender-neutral words from Gen-data as a part of training the MLPs and an extra category apart from female and male, i.e., neutral. Table 6 shows the percentage of neutral words misclassified in male class (I_1 - setting). It signifies that even for a gender-neutral word, BERT embeddings contain gender notion. Hence, simply removing the gender-specific words from the input sequence would not be a robust solution to tackle gender bias in downstream applications. However, the misclassification percentage decreases in case of BERT^{De}. Our proposed method does not need to avail any gender-specific information of an input word.

Conclusion

We studied gender bias induced by BERT in five downstream tasks. Using PCA, we identified orthogonal directions – defining a subspace – in BERT word embeddings that encode gender informative features. We then introduced an algorithm to identify fine-grained gender directions, i.e., 1-dimensional gender subspace. Omitting word vector components in such directions proved to be reducing gender bias in the downstream tasks. The methodology and Algorithm–1 can be adapted to study other social biases such as race, ethnicity, religion to make BERT's word embedding neutrals in respective biases.

Declarations

Informed Consent The authors did not receive support from any organization for the submitted work.

Ethical Standard This article does not contain any studies with human participants or animals performed by any of the authors.

Conflicts of Interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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